Taking the First Baby Steps in ns-3

Ns-3 is a network simulator for computer network simulation, used mainly for research and educational purposes. This article is the third in the series as we move forward in our quest to understand and use ns-3 by looking at abstracting classes and then working with our first script.

In the past two issues, we have dealt with many tools necessary to learn ns-3, which is a system of software libraries that work together to form a wonderful simulation tool. There are a lot of system libraries that are part of ns-3, and the user programs are written so that they can be linked to the core library. Let us first discuss the different classes available with ns-3 and the way they imitate and act as an abstraction of real world network entities. It may not always be feasible or possible to imitate a real world entity completely, with computer simulation. In such situations, we need to scale down things a bit, so that computers can handle the complex and confusing real world entities, without losing the essential characteristics of the particular entity being simulated. So an abstraction is a simplified version of an entity, with its essential characteristics. In order to set up a network, we need computer systems (nodes), applications running on top of these nodes, network devices for connectivity and connecting media. The section below discusses these real world entities and a few ns-3 abstracting classes imitating them.

A few built-in classes in ns-3

Node: In computer networking, node represents a host computer. In ns-3, the term node has a slightly different meaning. A C++ class, called *Node*, is used to implement nodes in ns-3 simulation. A node is a simplified model of a computing device in ns-3 and by creating an object of the *Node* class, you are actually creating a computing device in the simulation. For example, if there are 10 objects of the *Node* class in your simulation script, then you are simulating a network with 10 nodes. This is the power of ns-3. The developers of ns-3 have already finished the code required in your scripts to imitate a node in the real world; all you need to do now is call the *Node* class to create an object that imitates a real node.

Application: The behaviour of the application programs running on a real computer in a network is imitated by the *Application* class of ns-3. Using this class, a simple application can be installed on top of a node. A real computer node can have complex software like operating systems, system software and applications, but in ns-3, we content ourselves with simple applications, which are in no way comparable to real world software. Examples of applications are *BulkSendApplication*, *OnOffApplication*, etc. The *BulkSendApplication* is a simple traffic generator which will send data as fast as possible up to a maximum limit or until the application is stopped. The *OnOffApplication* is also a traffic generator and it follows an 'On/Off' pattern for sending